Aerospace Manufacturing Core
Study Guide

"AMCSG-305"

Blueprints & Documents

Documentation

Regulations

Text from Google AI

A Brief History of Aviation Regulation…

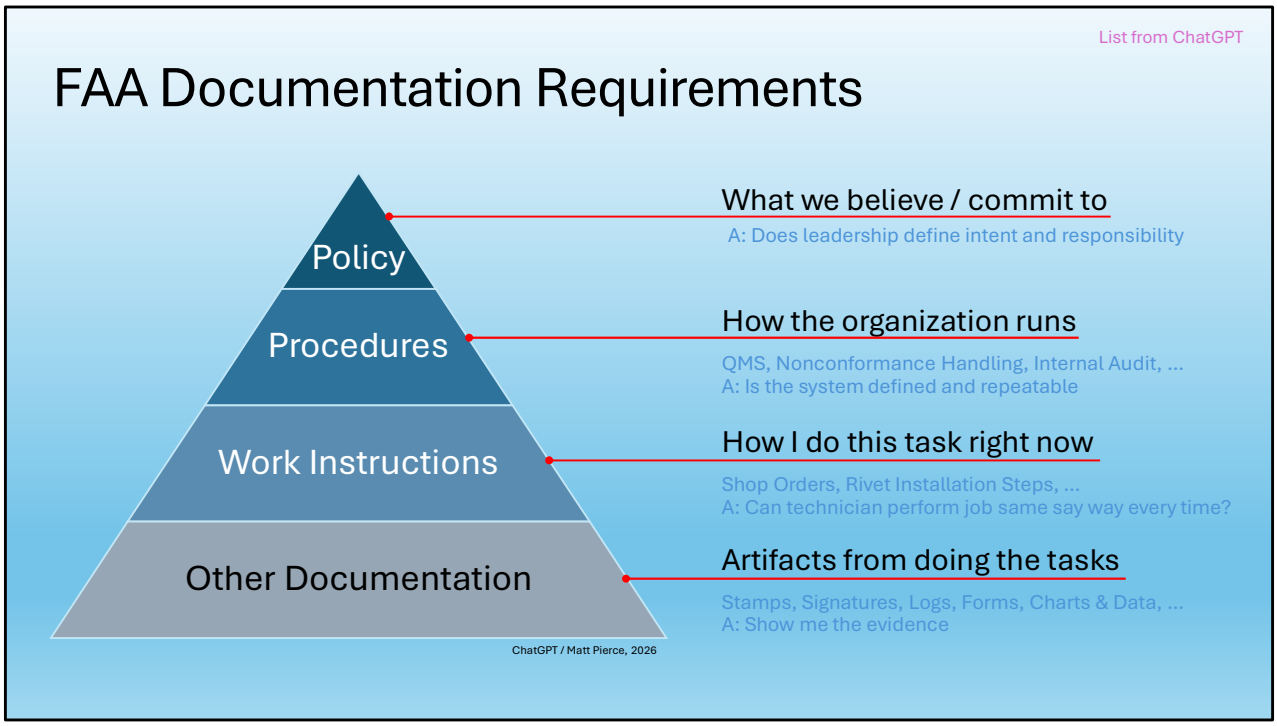
- **In the Beginning…**
  - "All fun and games"…only harm yourself if crash in an empty field
- **Air Commerce Act 1926**
  - Effectively transformed aviation into structured, safe, regulated industry
  - To: **Certify aircraft** as airworthy
  - To: **License pilots**
  - To: **Encourage growth** in the industry
- Federal Aviation Act 1958 (1956 collision over Grand Canyon)
  - Gave birth today's FAA
  - Absorbed former entity and duties, then expanded…
  - To: **Manage air navigation** (air traffic)
  - To: **Ensure safety** thru rules and standards

(PD) FAA.Gov

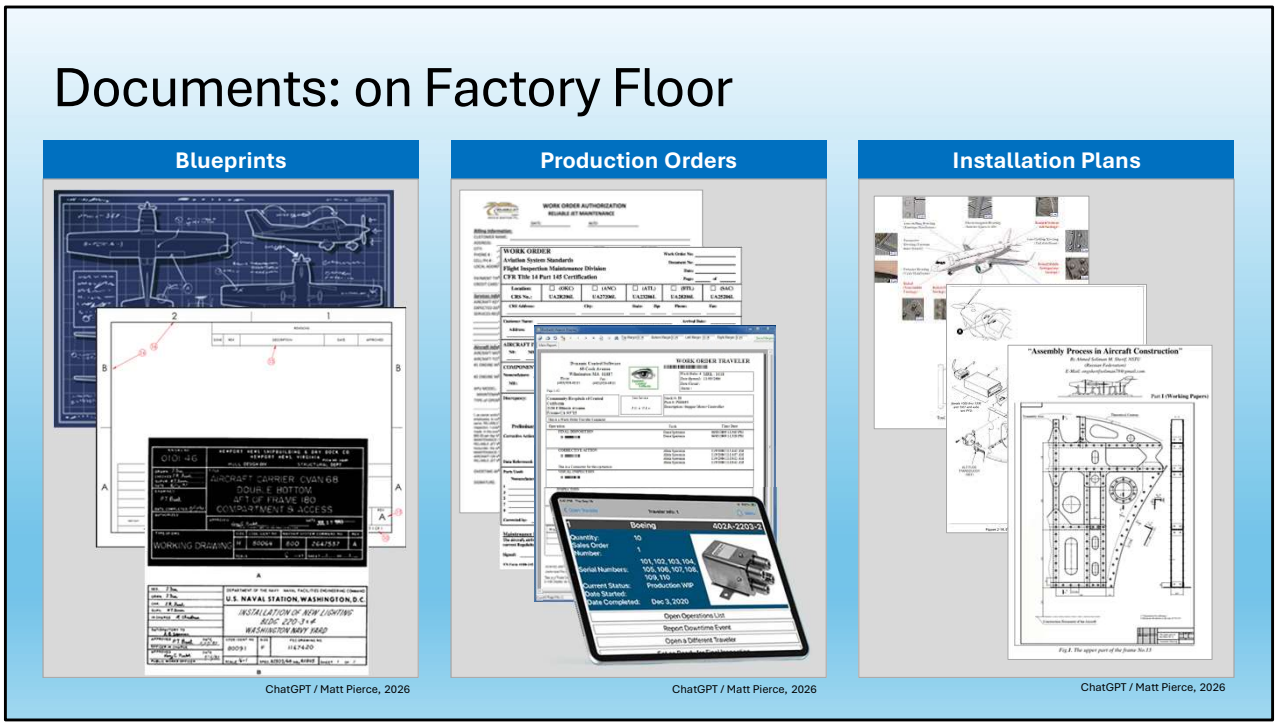
- Google: air commerce act 1926

Compliance

(Required Documents)



- ChatGPT: make an infographic about FAA mandated documentation types. In particular, compare policies vs. procedures vs. work instructions vs. other documentation types

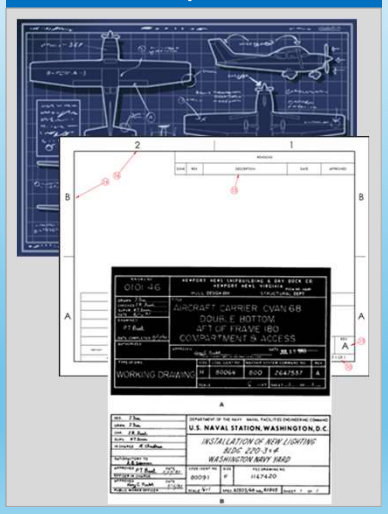


- ChatGPT: describe in an infographic the difference between blueprints, production orders, and installation plans.

Text from Google AI

Documents: Blueprints

Blueprints



ChatGPT / Matt Pierce, 2026

- **Blueprint Diagrams:** The Manufacturing Process
  1. **Detail Dwgs**: How to make a single part
  2. **Assembly Dwgs**: How to assemble 2 or more parts
  3. **Installation Dwgs**: How to final integrate assemblies
- **All have:**
  - Geometry and features / size and shape
  - Dimensions and tolerances
  - GD&T and datums
  - Notes and specifications / heat treating / machine finish / etc.
  - Revision authority
  - Part numbering, marking info, drawing number
  - Part list
- **Owned by:** Engineering
- **Used by:** Manufacturing, Quality, Tooling, and Suppliers

- Google: what is the difference between detail vs assembly vs installation for aerospace blueprints

- ChatGPT: describe in an infographic the difference between blueprints, production orders, and installation plans.

Text from ChatGPT

Documents: Production Orders

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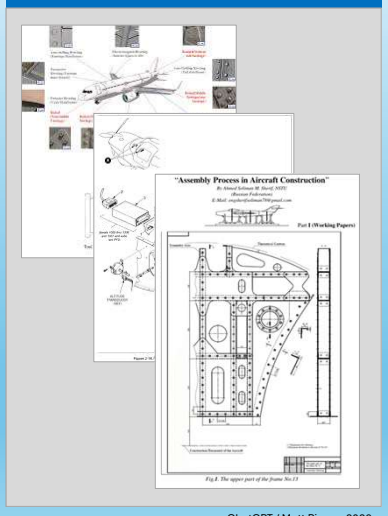
- **What:** Defines how to build and verify; step-by-step recipe with proof it was done.
- **Contains**:
  - Exact customer ordered “product configuration”
  - Drawing & spec references (which blueprints)
  - Operation sequence (what order to follow blueprints)
  - Required tooling
  - Inspection points
  - Sign-off / stamps / traceability
  - Serial or batch info
  - Work centers
- **Owned by:** Manufacturing / Planning
- **Used by:** Mechanics, Inspectors, and Planners

- ChatGPT: describe in an infographic the difference between blueprints, production orders, and installation plans.

Text from ChatGPT

Documents: Installation Plans (not Dwgs)

Installation Plans

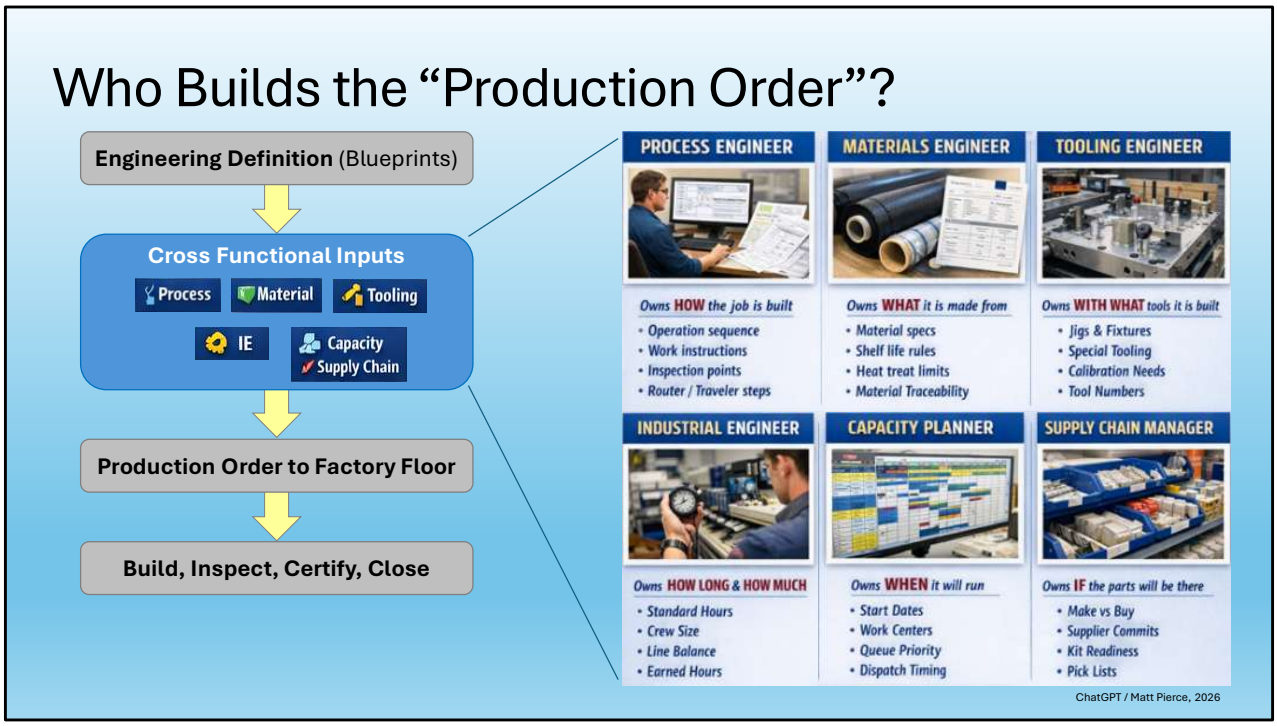


ChatGPT / Matt Pierce, 2026

- **What:** Defines where and how finished items fit; put this thing here, this way, with these connections.
- **Contains**:
  - Thousands of installation plans (a.k.a. work orders) are compiled for each plane
  - Location in aircraft (STA/WL/BL, etc.)
  - Orientation
  - Interfaces & mating parts
  - Fasteners & torque requirements
  - Routing / clearances
  - Access or rigging nots
- **Owned by:** Engineering (liaison / integration)
- **Used by:** Assembly & integration teams

- ChatGPT: describe in an infographic the difference between blueprints, production orders, and installation plans.

The “Production Order”



- ChatGPT: Make an infographic all the parties who create an aerospace "production order" and what they contribute. The parties are: process engineer, materials engineer, tooling engineer, industrial engineer, capacity planner, and supply chain managers
  + please make this into a single-page poster layout.

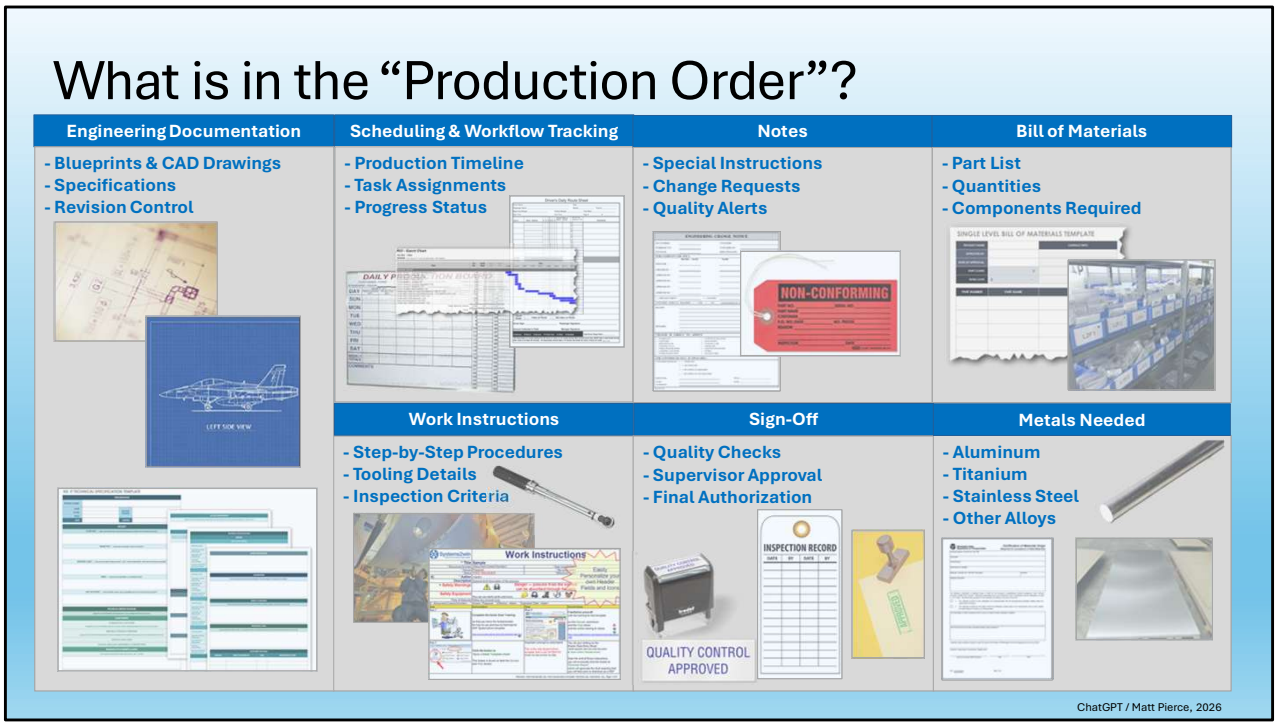


Image Credits:

- ChatGPT: Make an infographic showing all the important information and components of an aerospace "production order". Include: 1. "Scheduling and Workflow Tracking" 2. "Engineering Documentation" 3. "Notes" 4. "Bill of Materials" 5. "Metals Needed" 6. "Work Instructions" 7. "Sign Off“
  + Make this into a simpler power point training slide diagram
- ChatGPT: make six blueprint diagrams showing me all angles of an F18 fight jet from us government public domain sources. left side, plan (rear), top, front, right side, bottom
  + make the left side view only into a photograph
  + remove the background. jet only.
- ChatGPT: Make an infographic showing all the important information and components of an aerospace "production order".
  Include: "Scheduling and Workflow Tracking", "Engineering Documentation", "Notes", "Bill of Materials", "Metals Needed", "Work Instructions", and "Sign Off". Make it into a single poster format.

The “Installation Plan”

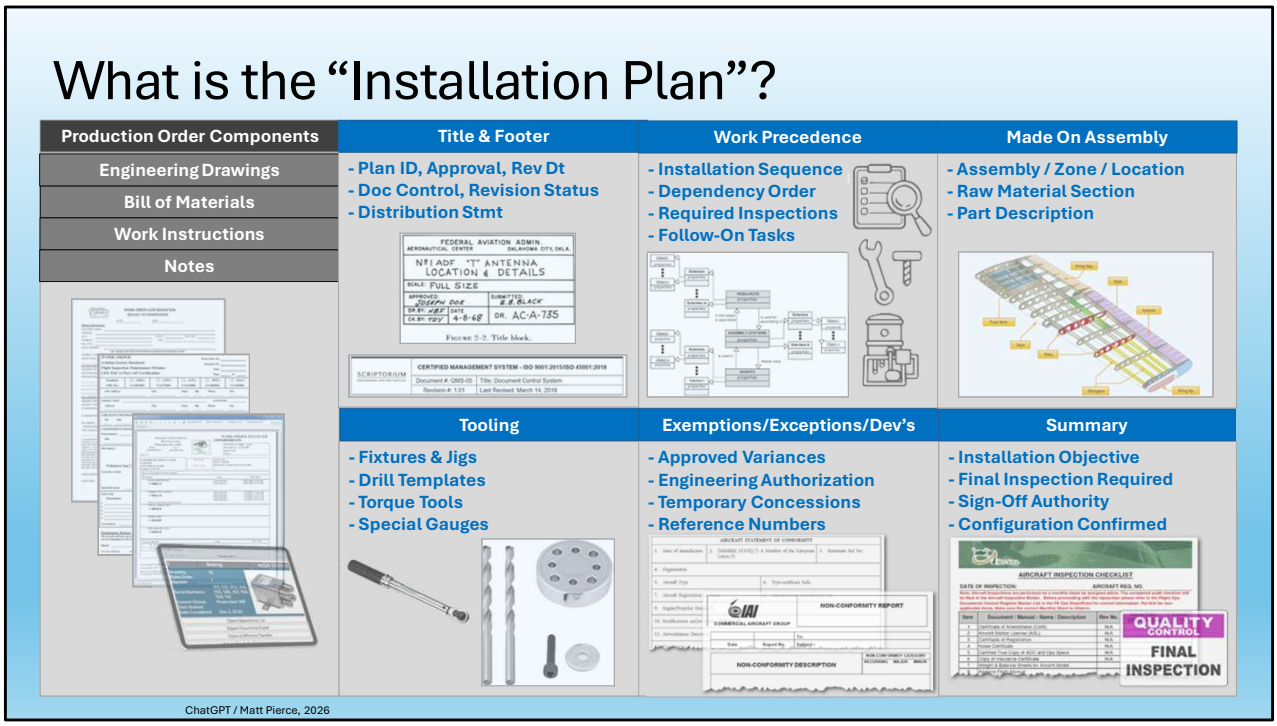
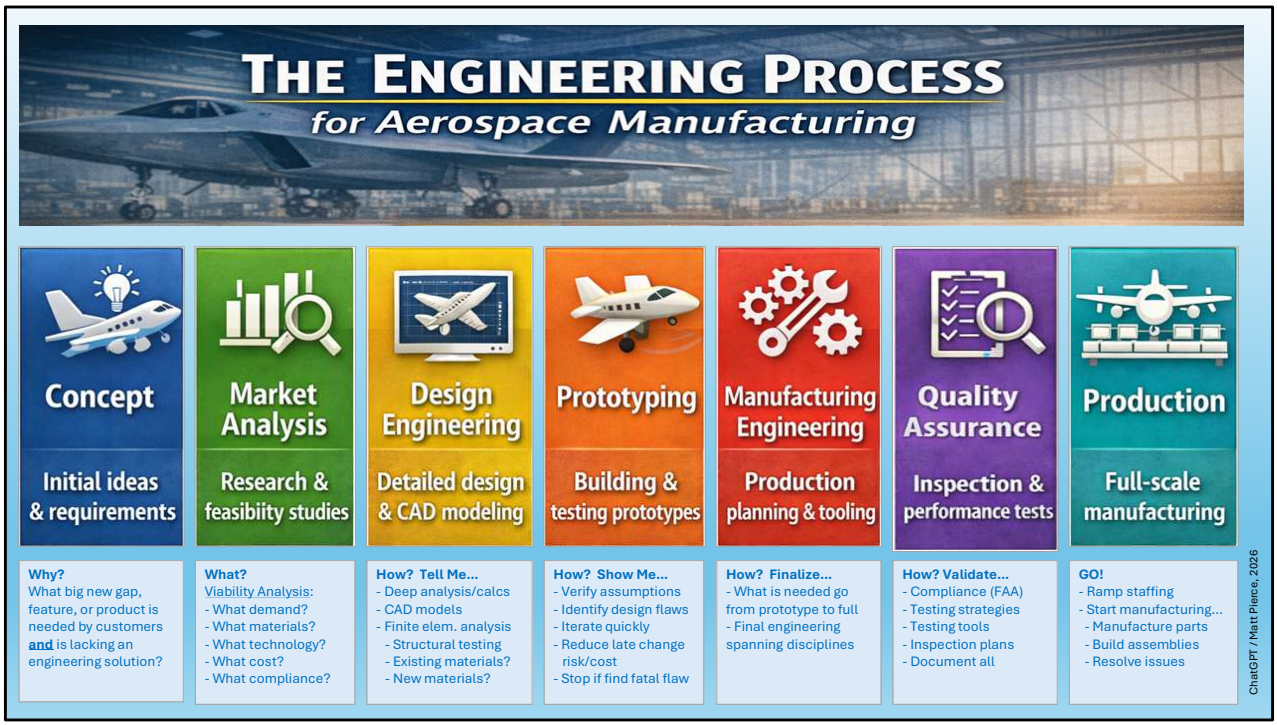


Image Credits

- ChatGPT: Make a diagram showing what an aerospace manufacturing "Installation Plan" is. Briefly explain each component: title, footer, bill of materials, work precedence, made on assembly, work instructions, engineering drawing, toilng, notes, exemptions/exceptions/deviations, and summary.
  + make this into a more visual icon-based infographic
  + an infographic simple diagram of all the information consolidated down.
  + a highly visual icon grid version (less text, more symbols)
  + Convert this into a clean printable poster graphic

The “Engineering Process”



- ChatGPT: make a diagram showing the phases of the "engineering process". For aerospace manufacturing. Phases are "concept", "market analysis", "design engineering", "prototyping", "manufacturing engineering", "quality assurance testing" and "production". Show all seven in order. Add a small description to each.
  + Make the phase graphics larger, fill up more space, larger font. All the text is great, do not change it. Make the background blank area smaller.
  + refactor the phases so they are rectangles instead of arrows.

Q&A

Practice Q&A's (Q1-Q8)

Q1: Document type where we commit to what we believe?

A: Policy

Q2: Document type where we define how the organization runs?

A: Procedures

Q3: Document type where we define how to do specific tasks (shop orders, rivet installation steps, etc.)?

A: Work Instructions

Q4: Document type where artifacts collect (stamps, logs, signatures, etc.)?

A: Other Documentation

Q5: Besides "Blueprints" and "Installation Plans", what document is on the factory floor?

A: Production Orders

Q6: Which document defines the manufacturing process, from how to make parts, to assembling parts, to final integration?

A: Blueprint Diagrams

Q7: Which document defines how to build and verify, step-by-step recipe with proof it was done?

A: Production Orders

Q8: Which document defines where and how finished items fit: put it here, this way, these connections?

A: Installation Plans

Practice Q&A's (Q9-Q16)

Q9: Y/N: Are "Installation Drawings" the same as "Installation Plans"?

A: No!
Installation drawings are a subset of blueprints. Installation Plans a separate superset beyond drawings.

Q10: Who builds the Production Order and what is the sequence?

A: Process Engineer > Materials Engineer
> Tooling Engineer > Industrial Engineer
> Capacity Planner / Supply Chain Manager

Q11: Y/N: The "Engineering Documentation" is in the Production Order?

A: Yes

Q12: Y/N: The "Scheduling & Workflow Tracking" are in the Production Order?

A: Yes

Q13: Y/N: The "Notes" are in the Production Order?

A: Yes

Q14: Y/N: The "Bill of Materials" is in the Production Order?

A: Yes

Q15: Y/N: The "Work Instructions" are in the Production Order?

A: Yes

Q16: Y/N: The "Sign-Off" is in the Production Order

A: Yes

Practice Q&A's (Q17-Q24)

Q17: Y/N: The "Metals Needed" is in the Production Order?

A: Yes

Q18: Y/N: The "Made On Assembly" is in the Production Order?

A: No. It is in the Installation Plan.

Q19: Y/N: The "Work Precedence" is in the Installation Plan?

A: Yes

Q20: Y/N: The "Tooling" is in the Installation Plan?

A: Yes

Q21: Y/N: The "Exemptions/Exceptions/ Deviations" is in the Installation Plan?

A: Yes

Q22: Part of Engineering Process to kick it all off with initial ideas and requirements?

A: Concept

Q23: Part of Engineering Process to research and do feasibility studies (demand? materials? ...)?

A: Market Analysis

Q24: Part of Engineering Process to perform detailed design and CAD modeling?

A: Design Engineering

Practice Q&A's (Q25-Q28)

Q25: Part of Engineering Process to build and test mockups?

A: Prototyping

Q26: Part of Engineering Process to plan tooling and production needs?

A: Manufacturing Engineering

Q27: Part of Engineering Process to plan inspection and performance tests?

A: Quality Assurance

Q28: Part of Engineering Process to put it all together into full scale manufacturing?

A: Production

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg